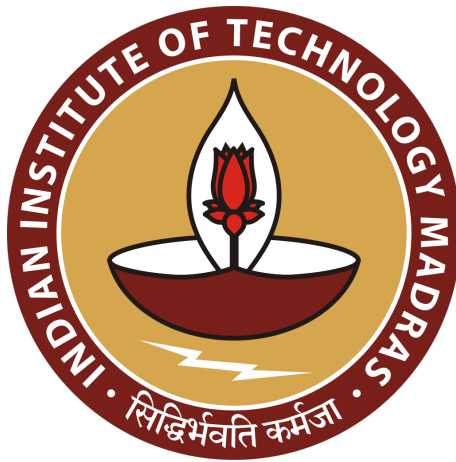


Lab Assignment - 9

Non Redundant Sequence Identification



BT3040 - Bioinformatics

Even Semester 2026

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Question 1

The code to calculate the pairwise Hamming Distance between the given sequences, and determine the closest pair of sequences is given below:

```
def Similarity_Hamming_Dist(sequences):
    composition, AAs = {}, "ACDEFGHIKLMNPQRSTVWY"

    for sequence_id in sequences.keys():
        composition[sequence_id] = {}

        for AA in AAs:
            composition[sequence_id][AA] = 0
        sequence = sequences[sequence_id]
        for AA in sequence:
            composition[sequence_id][AA] += 1

        for AA in AAs:
            composition[sequence_id][AA] = (composition[sequence_id][AA]/len(sequence))

    comparision_list = list(composition.keys())
    scores = []

    for n in range(len(comparision_list)):
        for m in range(n+1, len(comparision_list)):

            Hamming_Distance = 0
            sequence_1, sequence_2 = comparision_list[n], comparision_list[m]
            comparision = sequence_1 + " and " + sequence_2

            for AA in AAs:
                Hamming_Distance += abs(
                    composition[sequence_1][AA] - composition[sequence_2][AA]
                )
            scores.append((Hamming_Distance, comparision))

    scores.sort()
    print(f"The closest sequences in the above the with the least Hamming Distance  
are {scores[0][1]}, with a Hamming Distance of {scores[0][0]}")
    print("Ranking given below:")
    for n in range(len(scores)):
        print(f"{n+1})The Hamming Distance between {scores[n][1]} is:{scores[n][0]}")
```

The output of the program is shown below:

```
The closest sequences in the above the with the least Hamming Distance are  
sequence_1 and sequence_2, with a Hamming Distance of 0.665728476821192
```

```
=====
```

Ranking given below:

- 1)The Hamming Distance between sequence_1 and sequence_2 is: 0.665728476821192
- 2)The Hamming Distance between sequence_2 and sequence_3 is: 0.726632576075111
- 3)The Hamming Distance between sequence_1 and sequence_3 is: 0.8433544303797469

The code to calculate the pairwise Euclidian Distance between the given sequences, and determine the closest pair of sequences is given below:

```
def Similarity_Euclidian_Dist(sequences):
    composition, AAs = {}, "ACDEFGHIKLMNPQRSTVWY"

    for sequence_id in sequences.keys():
        composition[sequence_id] = {}
        for AA in AAs:
            composition[sequence_id][AA] = 0

        sequence = sequences[sequence_id]
        for AA in sequence:
            composition[sequence_id][AA] += 1

        for AA in AAs:
            composition[sequence_id][AA] = (composition[sequence_id][AA]/len(sequence))

    comparision_list = list(composition.keys())
    scores = []

    for n in range(len(comparision_list)):
        for m in range(n+1, len(comparision_list)):

            Euclidian_Distance = 0
            sequence_1, sequence_2 = comparision_list[n], comparision_list[m]
            comparision = sequence_1 + " and " + sequence_2

            for AA in AAs:
                Euclidian_Distance += (
                    (composition[sequence_1][AA] - composition[sequence_2][AA])**2
                )
            Euclidian_Distance = Euclidian_Distance**0.5
            scores.append((Euclidian_Distance, comparision))

    scores.sort()
    print(f"The closest sequences in the above with the least Euclidian Distance  
are {scores[0][1]}, with a Euclidian_Distance of {scores[0][0]}")
    print("Ranking given below:")
    for n in range(len(scores)):
        print(f"{n+1}) The Euclidian Distance between {scores[n][1]} is:  
{scores[n][0]}")
```

The output of the program is shown below:

```
The closest sequences in the above with the least Euclidian Distance are  
sequence_1 and sequence_2, with a Euclidian Distance of 0.20106216842153501  
=====
Ranking given below:
1) The Euclidian Distance between sequence_1 and sequence_2 is: 0.201062168421535
2) The Euclidian Distance between sequence_2 and sequence_3 is: 0.201129521072711
3) The Euclidian Distance between sequence_1 and sequence_3 is: 0.22086816691389
```

Question 2

The set of redundant beta-barrel membrane proteins was obtained from PDBTM as shown below:

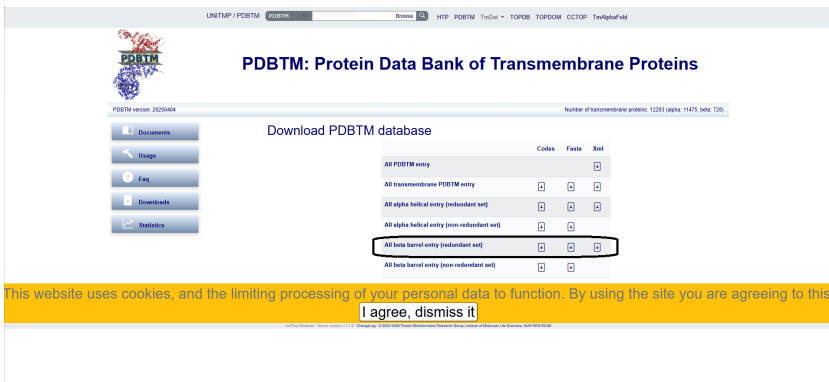


Figure 2.1: Obtaining sequence data of redundant beta-barrel membrane proteins

The commands listed below were used to run CD-HIT on the remote server, to obtain the clusters and representative sequences with sequence identities less than or equal to 40%, 50%, 75% and 90% respectively.

```
cd-hit -i Q2.fasta -o cd_hit_0.4_clustering -c 0.4 -n 2
cd-hit -i Q2.fasta -o cd_hit_0.5_clustering -c 0.5 -n 2
cd-hit -i Q2.fasta -o cd_hit_0.75_clustering -c 0.75 -n 2
cd-hit -i Q2.fasta -o cd_hit_0.9_clustering -c 0.9 -n 2
```

The results are summarized below:

S.No	Maximum Identity Threshold	Cluster Size (or) Number of Representative Sequences
1	40%	254
2	50%	281
3	75%	355
4	90%	397

Question 3

PISCES was used to obtain the set of non-redundant sequences with sequence identities of less than 20%, 30%, 40% and 50%. The steps to carry out the analysis are shown below:

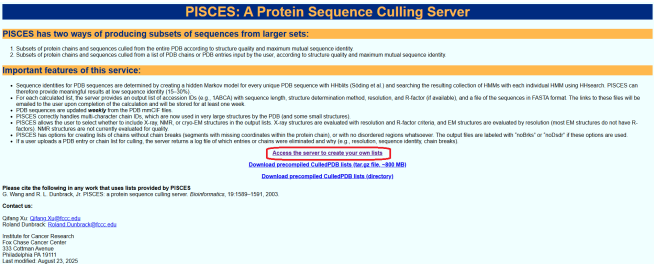


Figure 3.1: Creating own sequence lists on the PISCES server

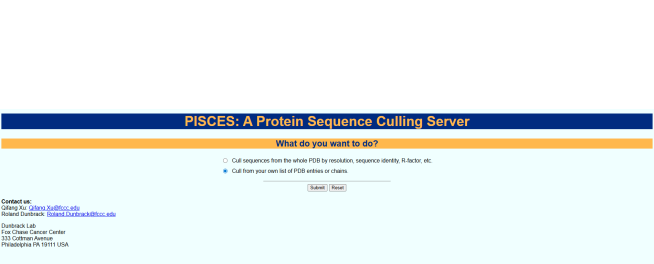


Figure 3.2: Custom PDB sequence filtering to obtain non-redundant sequences

NOTE: Default parameters were used for all parameters except sequence identity threshold.

PISCES: A Protein Sequence Culling Server

Step 1: Input PDB list

```

seq1_1  res1_1  met1_1  res1_2  met1_2  res1_3  met1_4  res1_5  a
seq1_6  res1_6  his1_1  res1_7  his1_2  res1_8  his1_3  res1_4  b
seq1_5  res1_5  val1_1  res1_6  val1_2  res1_7  val1_3  res1_4  c
seq1_4  res1_4  leu1_1  res1_5  leu1_2  res1_6  leu1_3  res1_4  d
seq1_3  res1_3  arg1_1  res1_4  arg1_2  res1_5  arg1_3  res1_4  e
seq1_2  res1_2  asp1_1  res1_3  asp1_2  res1_4  asp1_3  res1_4  f

```

Step 2: Choose your desired thresholds

Maximum pairwise percent sequence identity

Minimum resolution (X-ray and EM)

Maximum resolution (X-ray and EM)

Maximum E-value (X-ray only)

Minimum chain length

Maximum chain length

Include X-ray only?

☐ Yes ☐ No

Include cryo-EM entries?

☐ Yes ☐ No

Include NMR entries?

☐ Yes ☐ No

Include chains with chain breaks? (clinking "No" will reduce number of chains by ~50%)

☐ Yes ☐ No

Include chains with missing residues due to disorder? (clinking "No" will reduce number of chains by 50-70%)

☐ Yes ☐ No

Write file of pairwise sequence identities if returned?

☐ Yes ☐ No

Step 3: Submit your job

Figure 3.3: Querying for sequences with less than 20% sequence identity

[illegible]

Figure 3.5: Querying for sequences with less than 30% sequence identity

[illegible]

Figure 3.7: Querying for sequences with less than 40% sequence identity

[illegible]

Figure 3.9: Querying for sequences with less than 50% sequence identity

The results are compiled in the table below:

[illegible]

Figure 3.4: Results for sequences with less than 20% sequence identity

results from PISCES cullipbc_pc30_0_res0_0_0_2_len40-10000_R0.25_Xray_d0206_03_29_chains39

related_data@frcr.edu

Your threshold for calling the FGB
Resolution: 150 Å
B factor: 0.25
Sequence length: 81 - 10000
Sequence percentage identity: >= 30 %
X-ray entries: Included
EM entries: Excluded
MMDB entries: Excluded
Chains with chain breaks: Included
Chains with disorder: Included

Number of chains: 39

Download result files: https://ukfastk.org.uk/results/analysis/biochan/2067055/download.htm?ac=cullipbc_pc30_0_res0_0_0_2_len40-10000_R0.25_Xray_d0206_03_29_chains39

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Figure 3.6: Results for sequences with less than 30% sequence identity

results from PISCES cullipdb_pc40_0_res0_0_20_len40-10000_R0.25_Xray_d2026_03_29_chains42

relan.dunbrack@fccc.edu
View

Sun, Mar 26, 12:07 PM (2 days ago)

Your thresholds for culling the PDB

Resolution	-0.25 Å
R-factor	0.25
Sequence length	100-10000
Sequence percent identity	>= 40.0
X-ray sources	Included
EM sources	Excluded
NMR entries	Excluded
Chains with chain breaks	Included
Chains with disorder	Included

Number of chains: 42

Download results from: https://relaxrdb.fccc.edu/view/entries/281674/fd8a115c-417646-7a01f0-2026_03_29_chains42

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Figure 3.8: Results for sequences with less than 40% sequence identity

results from PISCES pulseq_c050_0_res0_0_20_len40-10000_R0_25_Xray_d2026_03_29_chains46

related.dunbrack@cs.fdu

10 min • 100% PROX • 100%

Sun, Mar 29, 10:10 PM (2 days ago) ☆ ☆

Your thresholds for cutting the PGB

Reads: 0.0 - 2.0

R-score: 0.25

Sequence length: 40 - 10000

Sequence percentage identity: <= 50.0

EM entries: Excluded

K-NN entries: Excluded

Chains with chain breaks: Included

Chains with disorder: Included

Number of chains: 46

Download result file: https://cloudstor.s3.amazonaws.com/28076154/f/external file: pulseq_c050_0_res0_0_20_len40-10000_R0_25_Xray_d2026_03_29_chains46.zip

Figure 3.10: Results for sequences with less than 50% sequence identity

S.No	Maximum Identity Threshold	Cluster Size (or) Number of Representative Sequences
1	20%	31
2	30%	39
3	40%	42
4	50%	46

Question 4

Comparing the results from both the tools, we clearly see that

- At 40% identity, 50% identity the total number of clusters obtained from **CD-HIT** is far greater than that obtained from **PISCES**.
- Additionally on increasing the identity threshold from 40% to 50% in **CD-HIT** the number of clusters increases from 254 to 281. However, in the case of **PISCES** the number sequences increases from 42 to just 46, when identity threshold increases from 40% to 50%.

This clearly indicates that the Hidden Markov Model used by PISCES, though it provides a broader range of sequence identity thresholds compared to CD-HIT, is more stringent in its grouping criteria and may yield fewer clusters at higher identity thresholds.

Question 5

The set of non-redundant beta barrel protein sequences was obtained from PDBTM in FASTA format as shown below:

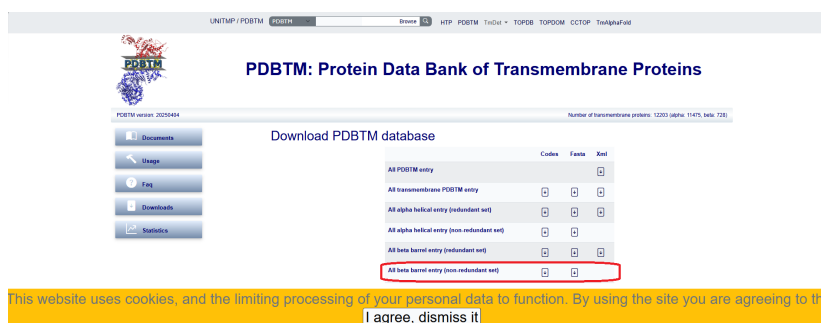


Figure 5.1: Obtaining sequence data of non-redundant beta-barrel membrane proteins

We observe that the non-redundant dataset of beta-barrel protein sequences obtained from PDBTM contains 253 entries. Comparing this with the results obtained using **CD-HIT** and **PISCES**, we note the following:

- **CD-HIT** yields 254 clusters (representative sequences), which closely matches the number of entries in non-redundant dataset of beta-barrel protein sequences obtained from PDBTM.
- **PISCES**, on the other hand, produces only 42 clusters (representative sequences), which deviates significantly from the dataset obtained from PDBTM.

This further reinforces that the Hidden Markov Model employed by **PISCES** for determining sequence identity is considerably more stringent, leading to increasingly conservative clustering as the identity threshold rises.

Link to the google drive containing the code and the result files from the different tools used in this assignment: <https://drive.google.com/drive/u/0/folders/1m9nC9oGx2GI3s1XrNVrwKPhnvYZ37Gr>

Annotations have been omitted from the submitted PDF for the sake of brevity. The fully annotated version of the code is available in the code folder within the attached Google Drive.
